

The Kalman Filter

A Complete Mathematical Reference

1. The Problem Setting

The Kalman filter addresses a fundamental problem in estimation: you have a system whose state evolves over time, you cannot observe the state directly, and your measurements of the state are corrupted by noise. How do you maintain the best possible estimate of the true state as measurements arrive?

The word “best” has a precise meaning here. The Kalman filter produces the **minimum mean squared error** (MMSE) estimate of the state — no other linear estimator can do better under the assumptions of the model. For linear-Gaussian systems, this estimate is also the maximum likelihood and maximum a posteriori estimate. The Kalman filter achieves all of these optimality properties simultaneously, which is why it has become one of the most widely used algorithms in engineering and science.

2. The State-Space Model

The Kalman filter assumes a **linear-Gaussian state-space model** consisting of two equations.

The **state transition equation** describes how the true (hidden) state x_t evolves from one time step to the next:

$$x_t = F x_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q)$$

Here F is the state transition matrix, w_t is process noise with covariance Q . In the simplest one-dimensional case (a scalar state like RSSI), $F = 1$ (the state is expected to persist) and Q is the process noise variance — how much the true state is expected to change between steps.

The **observation equation** describes how the noisy measurement y_t relates to the true state:

$$y_t = H x_t + v_t, \quad v_t \sim \mathcal{N}(0, R)$$

Here H is the observation matrix (in the scalar case, $H = 1$), and v_t is measurement noise with covariance R .

The filter assumes that process noise and measurement noise are independent of each other and of the initial state.

3. The Filter Equations

The Kalman filter operates in two alternating phases: **prediction** (project the current estimate forward in time) and **update** (incorporate the new measurement).

3.1 Prediction Step

Given the posterior estimate $\hat{x}_{t-1|t-1}$ and its covariance $P_{t-1|t-1}$ from the previous step, project forward to the current time:

$$\begin{aligned}\hat{x}_{t|t-1} &= F \hat{x}_{t-1|t-1} \quad \text{(predicted state)} \\ P_{t|t-1} &= F P_{t-1|t-1} F^T + Q \quad \text{(predicted covariance)}\end{aligned}$$

The predicted covariance $P_{t|t-1}$ is larger than $P_{t-1|t-1}$ by the amount Q — the filter acknowledges that it has become less certain about the state because the state may have changed since the last measurement.

3.2 Update Step

When a new measurement y_t arrives, update the predicted estimate:

$$\begin{aligned}K_t &= P_{t|t-1} H^T (H P_{t|t-1} H^T + R)^{-1} \quad \text{(Kalman gain)} \\ \hat{x}_{t|t} &= \hat{x}_{t|t-1} + K_t (y_t - H \hat{x}_{t|t-1}) \quad \text{(updated state)} \\ P_{t|t} &= (I - K_t H) P_{t|t-1} \quad \text{(updated covariance)}\end{aligned}$$

The term $(y_t - H \hat{x}_{t|t-1})$ is called the **innovation** — the part of the measurement that is new information, not predicted by the prior estimate. The Kalman gain K_t determines how much weight to give the innovation.

4. The Kalman Gain: Balancing Trust

The Kalman gain K_t is the heart of the filter, and its structure reveals exactly what the filter is doing. In the scalar case ($F = H = 1$):

$$K_t = \frac{P_{t|t-1}}{P_{t|t-1} + R}$$

This is a ratio between **predicted uncertainty** $P_{t|t-1}$ and **total uncertainty** $P_{t|t-1} + R$ (predicted uncertainty plus measurement noise). When $P_{t|t-1} \gg R$ — when the prior is very uncertain relative to the measurement noise — the gain approaches 1 and the filter trusts the measurement almost completely. When $P_{t|t-1} \ll R$ — when the prior is very confident relative to the measurement noise — the gain approaches 0 and the filter largely ignores the new measurement.

This is the correct behavior: the filter weights the prior and the measurement inversely proportional to their respective uncertainties. The Kalman filter is, in this sense, a principled interpolation between the prior prediction and the new observation.

5. Steady-State Behavior

Under mild conditions, the Kalman gain K_t and the covariance P_t converge to constant values as $t \rightarrow \infty$. This **steady-state Kalman filter** is simpler and computationally cheaper — you compute the steady-state gain K_∞ offline and use it for all subsequent updates.

In the scalar case, the steady-state gain satisfies:

$$K_\infty = \frac{Q + P_\infty}{Q + P_\infty + R} \quad \text{where} \quad P_\infty = \frac{Q}{2} + \sqrt{\frac{Q^2}{4} + QR}$$

For RSSI filtering with typical parameters ($Q = 10^{-5}$, $R = 2.0$), the steady-state gain is very small — the filter places most weight on its running estimate and only slightly adjusts for each new measurement. This produces the smooth, stable signal estimate that is useful for decision-making: it filters out the large measurement-to-measurement fluctuations in raw RSSI while still tracking genuine changes in signal strength.

6. Connection to Bayesian Inference

The Kalman filter is a special case of Bayesian filtering — sequential Bayesian inference for state-space models. At each step, the filter maintains a **belief state** $p(x_t \mid y_1, \dots, y_t)$ — the posterior distribution over the current state given all observations so far.

For linear-Gaussian models, this posterior is Gaussian at every step, fully characterized by its mean $\hat{x}_{t|t}$ and covariance $P_{t|t}$. The prediction and update equations are simply the Bayesian prediction and update applied to Gaussian distributions.

The prediction step applies the law of total probability: $p(x_t \mid y_{1:t-1}) = \int p(x_t \mid x_{t-1}, p(x_{t-1} \mid y_{1:t-1}), dx_{t-1}$

For linear-Gaussian dynamics, this convolution of Gaussians is itself Gaussian — the predicted mean and covariance follow directly from the linear transformation of the prior Gaussian.

The update step applies Bayes' theorem: $p(x_t \mid y_{1:t}) \propto p(y_t \mid x_t), p(x_t \mid y_{1:t-1})$

For Gaussian likelihood and Gaussian prior, the posterior is Gaussian, and its parameters are exactly the Kalman update equations. This connection to Bayesian inference means the Kalman filter is not just a heuristic smoothing algorithm — it is the exact solution to the Bayesian filtering problem under linear-Gaussian assumptions.

7. Relationship to the GP

The Kalman filter and the Gaussian Process are deeply connected. A GP with a stationary kernel can be expressed as an equivalent state-space model, which can then be filtered using the Kalman filter. This connection, developed by Hartikainen and Solin (2010), reduces GP inference from $O(n^3)$ to $O(n)$ for temporal GPs — a dramatic computational improvement for sequential data.

The key insight is that many stationary kernels — Matérn kernels in particular — correspond to the covariance functions of linear stochastic differential equations (SDEs). The solution to the SDE is a Markov process that can be filtered exactly with the Kalman filter. The Matérn $5/2$ kernel corresponds to an SDE whose state space has dimension 3, meaning you need a 3-dimensional Kalman filter to exactly replicate GP inference with this kernel.

This equivalence has both theoretical and practical importance. Theoretically, it unifies two seemingly different approaches to modeling noisy signals — GP regression and state-space filtering — into a single framework. Practically, it enables efficient temporal GP inference at scale, which is directly relevant for systems that must process long streams of sequential observations.

See also: Gaussian Process Regression; The Two-Timescale Bayesian Update